

MANAGEMENT PRESENTATION

CUSTOMER SATISFACTION ANALYSIS

Olist Brazilian E-Commerce Marketplace

PROBLEM OVERVIEW

Predict customer satisfaction and identify key drivers of dissatisfaction using Olist data.

GOAL

Predict customer satisfaction and identify the key drivers of dissatisfaction to support data-driven business decisions.

SCOPE

In Scope

- Olist order, delivery, seller and product data
- Data analysis and feature engineering
- ML model comparison and optimisation
- SHAP/LIME explainability
- Power BI dashboard and business recommendations

Out of Scope

- Real-time model deployment
- Integration with live Olist systems
- Analysis beyond the provided historical dataset
- Individual seller/carrier root-cause investigation

Business Questions

1. What drives customer dissatisfaction?

Identify the key factors influencing customer satisfaction, especially delivery delays and seller quality.

2. Can we predict dissatisfied customers?

Build a machine learning model to identify orders at risk of customer dissatisfaction.

3. How can Olist improve satisfaction?

Translate model findings into actions to reduce delivery delays, improve seller performance and proactively support at-risk orders.

DATA SUMMARY

What we worked with

- **Target:** 1 = satisfied, 0 = unsatisfied (review score ≥ 4).
- **Predictors:** Order, delivery, seller, product, payment and customer attributes.
- **Derived features:** delivery_delay_days, is_late, processing_time_days, freight_ratio, order_hour, seller_avg_score and frequency-based features.
- **Preprocessing:** Joined 9 Olist tables to order level, retained delivered orders with valid reviews, handled duplicates/missing values, translated categories, encoded categorical variables and applied imbalance handling.
- **Feature selection:** Reduced **25** → **14** features for the final XGBoost model.



KEY EDA INSIGHTS



Satisfaction by product_category_name_english:

product_category_name_english	count	satisfaction_rate
bed_bath_table	9090	74.26
health_beauty	8567	80.97
sports_leisure	7448	81.58
computers_accessories	6480	77.28
furniture_decor	6142	76.34
housewares	5646	79.97
watches_gifts	5436	77.74
telephony	4049	76.04
auto	3774	79.54
toys	3757	81.08

KEY EDA INSIGHTS

- Delivery is the strongest EDA signal: satisfaction is 82.79% for on-time orders vs 34.56% for late orders.
- Satisfaction varies by category: bed & bath table has the lowest rate among the major categories at 74.26%.
- Class imbalance: 78.93% satisfied vs 21.07% unsatisfied, making unsatisfied-order recall important for modelling.

category results include:

- Bed & bath table: **74.26%**
- Furniture/decor: **76.34%**
- Telephony: **76.04%**
- Computers/accessories: **77.28%**
- Watches/gifts: **77.74%**

MODEL COMPARISON SUMMARY

Baseline → Advanced → Tuned

Model	Accuracy	Macro F1	ROC-AUC	Recall (Unsatisfied)
Logistic Regression (Baseline)	0.7272	0.6262	0.6919	0.4921
Random Forest	0.8111	0.6442	0.7001	0.2997
LightGBM	0.7362	0.6353	0.7057	0.4985
XGBoost	0.7393	0.6372	0.7048	0.4958

Final Model: XGBoost

Selected based on the **best 5-fold CV ROC-AUC (0.7094)** and a more reasonable train–validation gap than Random Forest. After tuning and feature selection (**25 → 14 features**), ROC-AUC improved from **0.6919 to 0.7062 (+0.0143)** over the baseline.

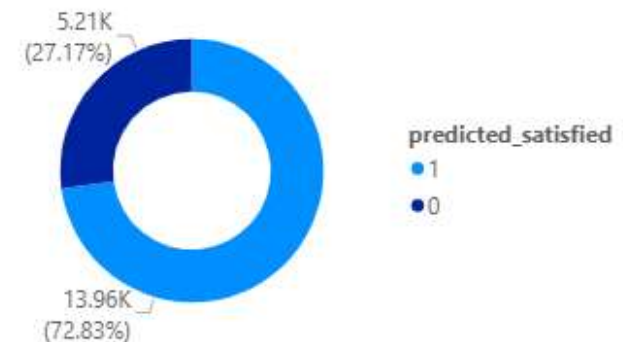
FINAL PREDICTIONS

From model to action

Final model: Tuned XGBoost

- Test ROC-AUC: **0.7062**
- Features reduced from **25** → **14**
- **19,165** test orders scored
- Each order receives a predicted satisfaction class and probability.
- At the selected **0.41 threshold**, unsatisfied recall reached **70.93%**.
- High-risk orders can be prioritised for proactive intervention.

Predicted Customer Satisfaction



Final XGBoost predictions were generated for 19,165 test orders. 72.83% were predicted satisfied and 27.17% predicted unsatisfied. At the selected 0.41 threshold, unsatisfied recall was 70.93%, enabling more dissatisfied customers to be identified for proactive action.

STRATEGIC VIEW — SWOT ANALYSIS

Mapped to the model's drivers

STRENGTHS

- Delivery delay is a clear and measurable driver of dissatisfaction.
- XGBoost provides explainable predictions using SHAP and LIME.
- Model identifies at-risk orders at the individual level.

WEAKNESSES

- Final ROC-AUC improvement over baseline is modest (**+0.0143**).
- Unsatisfied customers are a minority class (~**21%**).
- Random Forest showed strong overfitting.

OPPORTUNITIES

- Reduce delivery delays and improve delivery reliability.
- Improve seller quality and support low-performing sellers.
- Use prediction probabilities to proactively identify at-risk orders.

THREATS

- Historical patterns may not remain stable over time.
- Incorrect or incomplete delivery/review data can affect predictions.
- Optimising for unsatisfied recall involves a **precision trade-off**

Prioritise delivery performance and seller quality. Use the model to flag high-risk orders early, enabling proactive customer support and targeted seller interventions.

BUSINESS RECOMMENDATIONS

What to do?

1. Reduce delivery delays

Prioritise delayed orders and improve delivery reliability.

2. Improve seller performance

Monitor low-performing sellers and provide targeted support.

3. Proactively manage at-risk orders

Use model predictions to identify likely dissatisfied customers and intervene early.

4. Review high shipping costs

Investigate orders with high freight cost relative to item value.

Success Metrics

NORTH STAR:

- Customer Satisfaction Rate

OKRs:

- Reduce **late-order rate**
- Increase **satisfaction rate for late orders**
- Increase **recall of unsatisfied customers** through model-based intervention
- Improve performance of **low-rated sellers**